

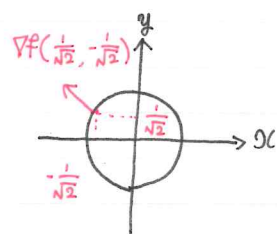
Chapter 3 Heat equation / Diffusion equation

 $f: \mathbb{R} \rightarrow \mathbb{R}^n$

$$\nabla f = \left(\frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_n} \right).$$

what is gradient?

Example 3.1: $f(x_1, x_2) = x_1^2 + x_2^2$
 $(x_1, x_2) \in \mathbb{R}^2$



$$\nabla f(x_1, x_2) = 2(x_1, x_2).$$

direction of $-\nabla f(x_1, x_2)$:most decreasing direction of fun. f at (x_1, x_2) .

<Derivation of PDE>

 $n \in \mathbb{N}$, $h > 0$: small, $t = nh$ given $f: \mathbb{R}^n \rightarrow \mathbb{R}$, we derive recurrence formula to decrease f .

$$\rightarrow x((n+1)h) = x(nh) - \gamma \nabla f(x(nh)).$$

define $\{x(nh)\}_{n=0}^{\infty}$

$$\Rightarrow \frac{x((n+1)h) - x(nh)}{h} = -\gamma \nabla f(x(nh)).$$

Letting $h \downarrow 0$ and $t = nh$, we have.

$$\frac{dx}{dt}(t) = -\gamma \nabla f(x(t)). \quad (3.2)$$

<Derivation of PDE>

$$h \nabla f = \frac{d}{d\varepsilon} \bigg|_{\varepsilon=0} f(x + \varepsilon h).$$

for any $h \in \mathbb{R}^n$

The previous lecture

$$\frac{d}{d\varepsilon} \bigg|_{\varepsilon=0} E(u + \varepsilon h) = \int_{\Omega} \nabla u(x) \cdot \nabla h(x) dx$$

for $h \in C_0^\infty(\Omega)$

By integration by parts.

$$\begin{aligned} \int_{\Omega} \nabla u(x) \cdot \nabla h(x) dx &= \int_{\Omega} (-\Delta u(x)) \cdot h(x) dx \\ &= (-\Delta u, h)_{L^2(\Omega)} \end{aligned}$$

Recall $f, g: \Omega \rightarrow \mathbb{R}$

From the idea to derive (3.2), we consider

$$\frac{\partial u}{\partial t}(x, t) = \gamma \Delta u(x, t) \quad \leftarrow \text{Heat / Diffusion eq.}$$

<Energy dissipation>

Prop 3.1: let $f: \mathbb{R}^n \rightarrow \mathbb{R}$ be a given smooth func.and let $r > 0$ be a positive numberLet $x(t): [0, \infty) \rightarrow \mathbb{R}^n$ be a sol. of

$$\frac{dx}{dt}(t) = -\gamma \nabla f(x(t)).$$

Then

$$\frac{d}{dt} f(x(t)) = -\gamma \underbrace{|\nabla f(x(t))|^2}_{\text{positive}} \leq 0$$

is hold



$$\begin{aligned}
 \text{pf. } \frac{d}{dt} f(x(t)) &= \nabla f(x(t)) \cdot \frac{dx}{dt}(t) \\
 &= \nabla f(x(t)) \cdot (-\gamma \nabla f(x(t))) \\
 &= -\gamma |\nabla f(x(t))|^2 \quad \square
 \end{aligned}$$

Prop 3.2: let $\gamma > 0$ be a positive constant let $u: \Omega \times [0, \infty) \rightarrow \mathbb{R}$ be a sol.

$$\begin{cases} \frac{\partial u}{\partial t}(x, t) = -\gamma \Delta u(x, t) & ; x \in \Omega, t > 0 \\ \nabla u(x, t) \cdot \nu(x) = 0 & , x \in \partial\Omega, t > 0 \end{cases}$$

Then

$$\begin{aligned}
 \frac{d}{dt} \left(\frac{1}{2} \int_{\Omega} |\nabla u(x, t)|^2 dx \right) \\
 = -\gamma \int_{\Omega} |\Delta u(x, t)|^2 dx \quad \square
 \end{aligned}$$

$$\begin{aligned}
 \text{pf. } \frac{d}{dt} \left(\frac{1}{2} \int_{\Omega} |\nabla u(x, t)|^2 dx \right) \\
 &= \frac{d}{dt} \left(\frac{1}{2} \int_{\Omega} \nabla u(x, t) \cdot \nabla u(x, t) dx \right) \\
 &= \int_{\Omega} \nabla u(x, t) \cdot \nabla \frac{\partial u}{\partial t}(x, t) dx \\
 &= \int_{\Omega} (-\Delta u(x, t)) \frac{\partial u}{\partial t}(x, t) dx \\
 &\quad \uparrow \text{Integrate by part} \\
 &= \int_{\Omega} (-\Delta u(x, t)) (\gamma \Delta u(x, t)) dx \\
 &= \gamma \int_{\Omega} |\Delta u(x, t)|^2 dx \quad \square
 \end{aligned}$$

<Other Properties>

Prop 3.3: let $\gamma > 0$ be a positive number (constant)

let $u: \Omega \times [0, \infty) \rightarrow \mathbb{R}$ be a sol of.

$$\begin{cases} \frac{\partial u}{\partial t}(x, t) = \gamma \Delta u(x, t) & , x \in \Omega, t > 0 \\ \nabla u(x, t) \cdot \nu(x) = 0 & , x \in \partial\Omega, t > 0 \end{cases}$$

Then

$$\frac{d}{dt} \int_{\Omega} u(x, t) dx = 0 \quad \leftarrow \int_{\Omega} u(x, t) dx \text{ doesn't change in time.}$$

is hold.

$$\begin{aligned}
 \text{pf. } \frac{d}{dt} \int_{\Omega} u(x, t) dx &= \int_{\Omega} \frac{\partial u}{\partial t}(x, t) dx \\
 &= \int_{\Omega} (\gamma \Delta u(x, t)) dx \\
 &\stackrel{\text{Eq.}}{=} \gamma \int_{\Omega} \text{div}(\nabla u(x, t)) dx \\
 &= \gamma \int_{\partial\Omega} \nabla u(x, t) \cdot \nu(x) d\Gamma(x) \\
 &\stackrel{\text{div thm}}{=} 0 \quad \square
 \end{aligned}$$

Prop 3.4: let $\gamma > 0$ be a positive constant

let $u: \Omega \times [0, \infty) \rightarrow \mathbb{R}$ be a positive sol of.

$$\begin{cases} \frac{\partial u}{\partial t}(x, t) = \gamma \Delta u(x, t) & , x \in \Omega, t > 0 \\ \nabla u(x, t) \cdot \nu(x) = 0 & , x \in \partial\Omega, t > 0 \end{cases}$$

let

$$\mathcal{H}[u] := \int_{\Omega} (\log u(x, t) - 1) u(x, t) dx$$

Then

$$\frac{d}{dt} \mathcal{H}[u] = -\gamma \int_{\Omega} |\nabla \log u(x, t)|^2 u(x, t) dx = 0.$$

is hold. $\square$

$\nearrow$ see note.